

Karthikadevi D

AI Engineer | Generative AI | RAG | Agentic AI

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years of experience spanning SQL/data workflows, database troubleshooting, Generative AI, Retrieval-Augmented Generation (RAG), and agentic AI applications. Experienced in building and optimizing LLM-based systems with a focus on RAG chunking strategies, latency optimization, retrieval quality, and production-oriented deployment. Passionate about building reliable, scalable, and low-latency AI solutions for real-world business problems.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS) | Aug 2024 - Present

AI Engineering, Generative AI & Data Solutions

- Started with SQL and data-focused responsibilities involving data integration, data cleaning, validation, and database support for production workflows.
- Investigated database-related production issues, analyzed failures, validated data, and supported troubleshooting and resolution activities.
- Contributed to a GenAI-powered database assistant providing query optimization suggestions, database troubleshooting guidance, and SOP-based knowledge assistance.
- Contributed to an agentic AI solution in the insurance domain involving RAG and LLM-based workflows, with a focus on RAG chunking strategy and end-to-end latency optimization.
- Worked with an AWS-based deployment environment involving Docker and Kubernetes.

TECHNICAL SKILLS

Generative AI: RAG, Agentic AI, LLMs, Prompt Engineering, Tool Calling, LangGraph, LangChain, Google ADK

Retrieval: ChromaDB, BM25, Hybrid Search, Reciprocal Rank Fusion (RRF), Sentence Transformers

Programming & Data: Python, SQL, SQLite

Development: FastAPI, Git/GitHub

Cloud & Deployment: AWS, Amazon Bedrock, Docker, Kubernetes

EDUCATION

B.E. - Electronics and Communication Engineering | VSB College of Engineering Technical Campus | 2020 - 2024 | **CGPA: 8.64**

Class XII | Kadir Mills Hr.Sec School | 2020 | **Percentage : 87.16 %**

Class X | VKN Matric School | 2018 | **Percentage : 91.8 %**

PROJECTS

Self-Healing Guardrailed RAG | Python, LangGraph, ChromaDB, BM25, RRF, Sentence Transformers, Groq

- Built a production-oriented RAG pipeline combining dense vector retrieval and BM25 keyword search using Reciprocal Rank Fusion.
- Implemented a LangGraph self-healing workflow with retrieval, generation, critic evaluation, query reformulation, retries, abstention, and fallback handling.
- Added prompt-injection detection, PII handling, schema validation, and YAML-driven input/output safety policies; benchmarked embedding and retrieval latency.

[GitHub: self-healing-guardrailed-rag](#)

Personal Mobile Voice AI Assistant | Google ADK, Gemini, Python, Android, SQLite

- Built a personal mobile voice assistant combining on-device speech recognition with a Google ADK/Gemini backend for natural-language interaction.
- Implemented contact lookup/call initiation, notes, reminders, general Q&A, Spotify media controls, Android notifications, text-to-speech, and automatic backend session recovery.
- Used a hybrid architecture: deterministic device actions execute locally while AI reasoning and tool selection are handled by the cloud backend.

[GitHub: personal-ai-assistant](#)

CERTIFICATIONS

AWS Certified Cloud Practitioner

AWS Certified AI Practitioner

Claude Developer Certification

Claude Architect Professional Certification